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Hybrid Recommendation Framework Combining Enhanced Density-Based Clustering and Transductive Support Vector Learning

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ABSTRACT: Recommendation Systems play a crucial role in improving business strategies and providing customers or users with the choice to opt for a product or service. Given the growth and power of information analytics tools and the inclination of data mining techniques, a wide range of recommender systems, particularly for web-based users, have evolved over the past decade. The conventional "One model – Fit for all" model fails in most cases due to the common fact that not all users have the same interests, and has paved the way for tailor-made recommender systems. It is hence highly important to provide a personalized framework that can accumulate the specific interests of the users. Web data are normally highly sparse in nature, and formulating a personalized user profile was a difficult task. The previous methods used user user-based CF method to address the former issue, while Extended Jaccard Similarity (EJS) was introduced to address the latter. Although these methods were found to be effective, there was scope for improvement in the Query processing time and accuracy. To solve this problem, the proposed system designs an Enhanced DBSCAN and TSVM-based hybrid recommendation scheme. In the proposed system, the user's interests and profile are analyzed by mining the web logs. After de-noising, user profile clustering is done through an improved DB scan, where Anarchic Society Optimization is used for parameter fine-tuning. The sequential pattern of the users has evolved based on the GSP approach. The user interests are then evaluated using the TSVM, which makes use of the weighted mean for the calculation of the trust value of the users. Lastly, the personalized recommendations are generated based on the clicks, selections, and browsing patterns. The proposed method is applied to MOVIE LENS data and found to perform better than the previous methods and other state-of-the-art recommendation models in terms of MAE, Speed, and Accuracy.

KEYWORDS: Recommendation Systems, DBSCAN, Support Vector Machines (SVM), Anarchic Society Optimization (ASO), Extended Jaccard Similarity (EJS).

1 Introduction

In the digital era, with the rise in online web-based services, recommender systems play a crucial role in day-to-day life. Recommender systems touch every aspect of the online world, starting from e-commerce to online advertisements that target the interests of a specific type of user. Generically, recommender systems are the methodologies or the algorithms that aim to suggest the most relevant items or services to a particular user (ex, the movies to watch, the books to read, to choose a product, etc.) [1-2]. The recommender systems have evolved as a critical parameter in many industries, as they stand as vital in business expansion and acquisition of customers [3-4].

Traditionally, recommender systems are categorized into collaborative and content-based filtering approaches [5-6]. The usage of the Hybrid approach has become more popular among researchers who combine the filtering approaches for increased performance. There are always many hindrances in combining these filter techniques [7-8]. Even the same type of techniques, when tried to hybridize face issues. Hybrid approaches are possible in many ways: by generating the recommendations through collaborative and content-based filtering and then combining the results, by incorporating the features of content-based filtering in collaborative filtering and vice versa, or by a unified approach that combines both methods. Various studies have proven that the hybrid approach outperforms better than the individual approaches [9]. These hybrid approaches are also proven to solve major issues such as data sparsity and user profile clustering. In short, the hybrid approach solves the bottleneck in knowledge engineering in various knowledge-based systems [10-11].

The previous methods used user user-based CF method to address the former issue, while Extended Jaccard Similarity (EJS) was introduced to address the latter. Although these methods were found to be effective, there was scope for improvement in the Query processing time and accuracy. To solve this problem, the proposed system designs an Enhanced DBSCAN and TSVM-based hybrid recommendation scheme. In the proposed system, the user's interests and profile are analyzed by mining the web logs. After de-noising, user profile clustering is done through an improved DB scan, where Anarchic Society Optimization is used for parameter fine-tuning. The sequential pattern of the users has evolved based on the GSP approach. The user interests are then evaluated using the TSVM, which makes use of the weighted mean for the calculation of the trust value of the users. Lastly, the personalized recommendations are generated based on the clicks, selections, and browsing patterns. The proposed method is applied to MOVIE LENS data and found to perform better than the previous methods and other state-of-the-art recommendation models in terms of MAE, Speed, and Accuracy.

2 Literature Survey

Wu-Yang et al (2019) proposed a methodology that combines Singular Value Decomposition (SVD) and Hybrid Collaborative Filtering for an effective and optimized solution for the data sparsity issue. The SVD is used for bringing in a reduction in user-view matrices that are obtained through web usage mining [12]. Both of the low-ranked matrices are then employed for generating item-based and user-oriented predictions. An automated framework for building an automatic system in a real-time scenario is proposed. The recommendation engine performs in an online, real-time scenario. The experimental results through the MOVIE LENS data prove the efficiency of the proposed method.

Hussein et al (2019) proposed the novel Hybrid framework, which builds complex and context-aware solutions in online recommendation. The proposed method is based on the concept of dynamic conceptualization [13]. The proposed framework gives a range of algorithms for producing a set of recommendations or also known as templates, to combine various methods into hybrid recommenders. The proposed framework was also meant for integrating the existing user or product information from outside sources, such as social networks, MovieLens, etc. The hybrid method combines the extraordinary characteristics of state-of-the-art methods and Collaborative filtering methods. The evaluation is based on three conceptual foundations. The results indicate that the proposed method was efficient in terms of accuracy by reducing the loss and Mean Square error value.

Shreya et al (2019) improved the quality of the Movie recommendation system through a Hybrid approach that combines both the CF models using SVM as a classifier. A genetic model is also proposed, and the comparative result showed significant improvement in terms of quality, scalability, and accuracy of the Movie recommendation [14].

G M Sarene et al (2020) investigated that the performance of a recommender system is more affected due to over- and above specifications, data sparsity, and scalability. In order to mitigate some of the major negative effects, a hybrid system called the Relevance-based Recommender was introduced [15]. The proposed method exploits the individual measures in predictive relevance that are computed for every user for every instance of interest, which ensures high precision and recall values. Experimental results show that the advantages which are introduced through this recommender system have potential suggestions.

J. Grivolla et al (2021) proposed a hybrid recommender system that integrates the demographics of users and the characteristics of the item set interactions [16]. The proposed system is then evaluated on the MOVIE Lens data, which includes gene information and the user data, which operates to provide a tailor-made discount to the customers. The proposed method included additional items and more information about the users, which has more information and impact on the quality of information, specifically where less interaction between the user profiles is available.

Palit R et al (2020) provided a basic recommender system and how to implement it using the content and collaborative filter methods [17]. The proposed model takes the matrix of user ratings of movies as the top inputs, and recommendations are generated. The proposed method also discusses the hybrid or mixture of recommender models for effective recommendations. The experimental results show that the proposed method outperforms other state-of-the-art methods.

3 Proposed Methodology

The Proposed methodology is shown in Figure 3.1

3.1 Data preprocessing

The system distinguishes various patterns of user style and behaviour during testing the browsing styles of users and taking out their server blogs. Normally, the log file might include various unnecessary behaviors in extremely short periods of time. Unfortunately, these behaviors might happen numerous times in a similar pattern. One more problem is that, in one pattern, it might include simply an action. These are between the noises to be removed from the beta version of sequential datasets. The noise finding and elimination steps are extremely significant since they be able to modify the result of information examination in the future.

Algorithm 1: Denoised based on one source de-noising

1. Given user u, content m, trial t=1,2,3, and > 0
2. Rut =$\{r_1, \dots, r_k\}(ratings\ \neq 0\ for\ user\ u\ in\ one\ trial)$
3. Rut' =$\{r_1, \dots, r_k\}(ratings\ \neq 0\ for\ user\ u\ in\ different\ trial)$
4. Being 0 the value of the non-rated item
5. for $r_i \in R_{ut}\ do$
6. if 0< $|R_{ut}(m) - R_{ut'}(m) > \gamma$ then
7. $R_{ut}(m)$=0(SD: we delete rating)
8. $else\ if\ 0 < |R_{ut}(m) - R_{ut'}(m)| > \gamma$ (MD: ratings in both trials differ by less than a given threshold) then
9. $R_{ut}(m) = M(R_{ut}(m), R_{ut'}(m))$
10. else

11. $R_{ut}(m) = R_{ut'}(m)$
12. end if
13. end for

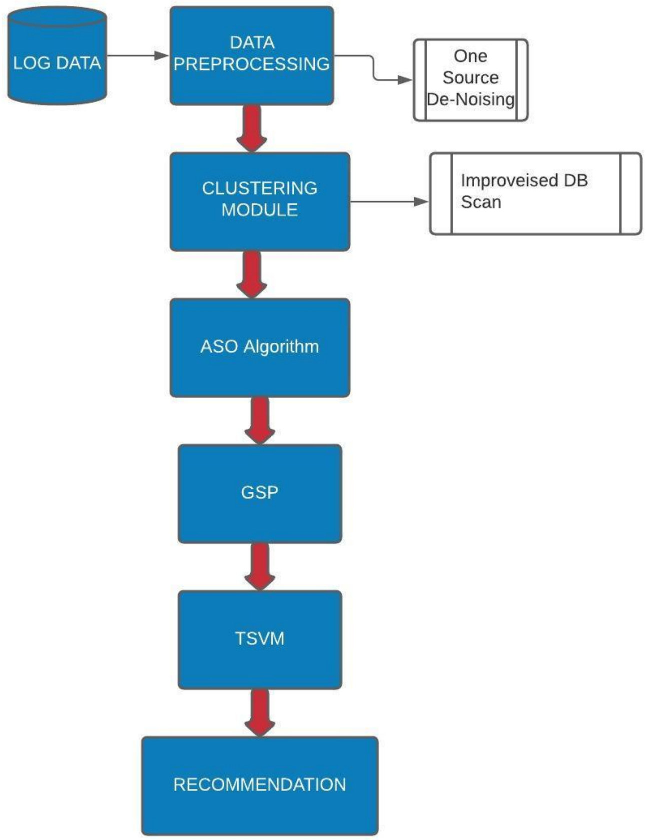

Figure 1: Proposed Methodology

In this work, if each one's rating differs by more than a specified threshold, there is a Strong Disagreement (SD) condition, and the rating is removed from the present dataset. At last, if at least two ratings are different by means of less than there is a Mild Disagreement (MD) condition.

3.2 Clustering Module

Recommendations should not be designed for the entire pool of users, but still for users by means of related specific interests; their ability to handle a task is able to differ appropriately toward variations in their information level. In the proposed work, a clustering technique is carried out as an initial step toward grouping users based on their browsing styles of the user. These groups are used to recognize logical choices in common sequences of user behaviors. Depending on the various patterns of browsing style and users' behaviors, the clustering is carried out via the use of the DBSCAN clustering algorithm.

DBSCAN is used to group similar kinds of user profiles. The DBSCAN method usually observes the user groups as dense regions of objects in the information space, with the purpose are dividing by means of regions of low-density objects. It generally has two parameters, such as Eps and MinPts, as inputs for grouping similar Users. From the following description, it simply requires finding out every one of the maximal densities connected spaces to cluster the users in a search space. These spaces are known as groups. Every user's data not contained in any group is measured as noise and be able to be ignored from the original Users. It makes use of the idea of density, such as reachability and connectivity.

3.2.1 Density Reachability

An users' data at a point "p" is said toward be density reachable from a user data at a point "q" if point "p" is in ε distance from Users data at a point "q" and "q" has sufficient number of user data at points in its neighbors who are inside distance ε.

3.2.2 Density Connectivity

Users' data at a point "p" and "q" are said in the direction of is density connected if there exist a users' data at a point "r" which has sufficient number of Users' data at a point in its neighbors and both the Users' data at a points "p" and "q" are in the ε distance. This is sequence progression. So, if "q" is neighbor of Users' data at a point "r", "r" is neighbor of Users' data at a point "s", "s" is neighbor of Users' data at a point "t" which in turn is neighbor of Users' data at a point "p" implies with the purpose of Users' data at a point "q" is neighbor of Users' data at a point "p".

Calculating the Parameters ε and Min Pts,

The dynamic steps permit two input Users' data points; the automatically created parameters differ in the direction of various Users' data points.

1. The distance between two Users' data points

The distance d (a, b) function measures the similarity of two Users' data points.

$$\mathrm{d}\ (\mathrm{a,b}) \ = \ \mathrm{q}\sqrt{(\mu_1|X_{a1} - X_{b1}\ |^{q} + \mu_2\ |X_{a2} - X_{b2}|^{q} + \cdots + \mu_p\left|X_{ap} - X_{bp}\right|^{q}} \quad (1)$$

Where, a = ($x_{a1}, x_{a2}, \ldots, x_{ap}$) and b = ($x_{b1}, x_{b2}, \ldots, x_{bp}$) are two p-dimensional Users' data points, and q is a positive integer.

2. Automatic parameters creation dynamic method

The proposed DBSCAN algorithm discovers groups inside varied densities by creating multiple pairs of ε and MinPts routinely.

Algorithm 1:

Let X = {$x_1, x_2, x_3 \ldots x_n$} be a set of Users' data points (users). DBSCAN requires three parameters: ε (eps), similarity, and the smallest number of Users' data points (users) necessary to form a cluster (minPts).

Step 1: Start with an arbitrary starting Users data point with the purpose of has not been visited

Step 2: Extract the neighborhood of this leaner's data points using ε.

Step 3: If there is similar browsing style of user around this Users data points (user) then clustering process starts and Users data points is marked as visited

Step 4: else Users data points is labeled as noise

Step 5: if a user's data points (user) is found to be a part of the cluster then its neighborhood is also the part of the group and the above procedure from step 2 is repeated for all ε neighborhood Users data points. This is repeated in anticipation of each and every one Users' data points (users) in the group are found.

Step 6: A new visited Users' data points (user) is retrieved and processed, leading in the direction of the discovery of a further cluster.

Step 7: This process continues until each and every one Users data points (users) are marked as visited.

The major issue with this clustering is that it needs to tune its two parameters ε and minPts. To solve this issue, these two parameters are tuned via the use of the Anarchic Society Optimization (ASO) algorithm [18]. Anarchic Society Optimization (ASO) is a new optimization method from swarm intelligence. Initially, several community parts are chosen randomly inside the search space for tuning of two parameters, ε and minPts. Then, the clustering accuracy of every part is calculated depending on the two parameters ε and minPts. According to the computed fitness value and comparison with X^*(k), $P_i^{best}$and G^{best}, the movement arrangement and a new value of the part will be found. After a sufficient number of iterations, at least one of the parts' determinations reaches the optimal position of the cluster parameters.

Algorithm 2: ASO Algorithm

Input: N parts

Output: Optimal parameters ε and minPts.

1. Initialize the N parts, part position and set of iterations counter I=0
2. Generate M initial solutions and evaluate their fitness values
3. While (termination criteria are not satisfied)
4. Determining the values of X^*(k), P_i^{best} and, G^{best}
5. For i=1:M
6. Computing FI_i (k)
7. If FI_i (k) is less than threshold X_i (k), then X_i (k) moves towards X^*(k)
8. Else X_i (k) moves towards a random part
9. End if
10. End for
11. For i=1:M
12. 12.Computing EI_i (k)
13. 3.If EI_i (k) is less than threshold X_i (k), then X_i (k) moves towards G^{best}
14. 14.Else X_i (k) moves towards a random part
15. 15.End if & End for
16. 17.For i=1:M
17. 18.Computing II_i (k)
18. 19.If II_i (k) is less than threshold X_i (k), then X_i (k) moves towards P_i^{best}
19. 20.Else X_i (k) moves towards a random part
20. 21.End if & End for
21. 23.For i=1:M
22. 24. Updating the position by combining the movement arrangement
23. Calculate the fitness values
24. End for & End While while
25. Report the best solution

26. End

3.3 Generalized Sequential Pattern (GSP)

Users with different browsing styles have various arrangements of rehashed succession. Consequently, students were grouped based on their styles of the students and afterward personal conduct standards were found intended for each student through the utilization of the Generalized Sequential Pattern algorithm (GPS) [19]. The large GPS is proposed for comprehending continuous example mining issues. One route toward making use of the level-wise model is to initially find every single one of the continuous students in a style which is level-wise manner. This essentially accounts for the checking of events of every last one singleton part in the database. In this way, non-frequent items are expelled to channel the exchanges. At the end of the work, each exchange incorporates just the incessant parts it initially contained. This altered database is then given as a contribution toward the GSP calculation. This progression needs one to disregard the entire database. Various disregards of the database are finished by methods for the GSP Algorithm. In the primary pass, every single one particular thing (1 sequence) is tallied. Accordingly, from the found continuous things, a place of candidate 2-sequences is made, and in this manner, pass is made to perceive their recurrence. These successive 2-groupings are utilized to make the candidate 3-sequences, and this calculation is repeated until the point that no more frequent sequences are found. The calculation incorporates two noteworthy advances.

3.3.1 Candidate Creation

Specified the set of frequent (i-1) and their frequent sequences Fn(i-1), the candidates used for the next pass are created by combining Fn(i-1) by means of itself. As a minimum, one of whose subsequences is not frequent is removed by means of the pruning step.

3.3.2 Pruning phase

A pruning step removes some sequence at least one of whose series is not frequent.

3.3.3 Support Counting

Frequently, a hash tree–based search is proposed for capable support counting. Subsequently lastly non-non-maximal frequent sequences are removed.

Algorithm

1. Fn1 = the set of frequent 1-sequence
2. i=2,
3. do while Fn(ii-1)!=Null;
4. Generate candidates sets Cni (set of candidates i sequences);
5. For all i/p sequences Q in the database D do
6. Raise count of all a in Cni if Q supports a Fni={a ∈ Cni such that its frequency exceeds the threshold} i=i+1;
7. Result =Set of all recurrent sequences is the union of all Fnis
8. End do
9. END

3.4 TSVM

To know the Users’ data, the transductive support vector machine is proposed in this work. Depending on the percentage ideal answer, the Users’ rating is able to be explained, which is completed in the direction of dividing the clustered users depending on the ratings of these frequent sequences [20]. TSVM is generally a continuous classifier that regularly finds the best hyperplane in the user's space with a transductive procedure, with the purpose of including unlabeled samples in the training step.

The browsing step of the TSVM can be designed as an optimization issue and described as follows:

$$J(w, \xi, \xi^*)=\frac{1}{2}||w||^2+C\sum_{i=1}^{l}\xi_i + C^*\sum_{j=1}^{d}\xi_j^* \quad (2)$$

Will be focused on,

$$y_i(\varphi X_i).w+b)\geq 1-\xi_i,\ \xi_i \geq 0; i = 1,2,\dots.l \quad (3)$$

$$y_j(\varphi X_j).w+b)\geq 1-\xi_j^*,\ \xi_j^* \geq 0; i = 1,2,\dots.d \quad (4)$$

Where C and C^*- user mentioned a penalty value of together the training and the transductive samples ξ_i, and ξ_j^* - random variables and d - total number of transductive samples. Training the TSVM relates in the direction of handling the above-mentioned issues. Lastly, the conclusion function of the TSVM with Lagrange multipliers α_i and α_j^* are described as follows,

$$f(x)= \text{sgn}\left[\left[\sum_{i=1}^{l} y_i\alpha_i k(x,x_i) + \sum_{j=1}^{d} y_j^*\alpha_j^* k(x,x_j^*+b)\right]\right] \quad (5)$$

If two Users were calculated by the system by means of the parallel rating designed for a comparable navigational series, then they are supposed to be similar. Recommendation procedure is able to be performed depending on these browsing styles, which is performed based on the Personalized Browsing Approach (PLA).

3.5 Recommendation process

The student preferences and browsing behavior be able to be described by means of the subsequent three ways:

1. Clicks: This step is described as a small listing of material.
2. Selection: This step is described as the material chosen and added to the cart.
3. Browsing: This step is described as reading the material

All of the above steps are used to recognize the Users' comparative preferences LP_{ij} used for every material referred from the information sources. The formula used to find the LP_{ij} is described as follows:

$$LP_{ij} = \frac{LP_{ij}^c - Min_{1\leq j\leq |M|}(LP_{ij}^c)}{Max_{1\leq j\leq |M|}\left(LP_{ij}^c\right) - Min_{1\leq j\leq |M|}(LP_{ij}^c)}$$

$$\frac{LP_{ij}^s - Min_{1\leq j\leq |M|}(LP_{ij}^s)}{Max_{1\leq j\leq |M|}\left(LP_{ij}^s\right) - Min_{1\leq j\leq |M|}(LP_{ij}^s)} + \frac{LP_{ij}^l - Min_{1\leq j\leq |M|}(LP_{ij}^l)}{Max_{1\leq j\leq |M|}\left(LP_{ij}^l\right) - Min_{1\leq j\leq |M|}(LP_{ij}^l)} \quad (6)$$

LP_{ij}^c, LP_{ij}^s and LP_{ij}^l describes the number of references toward material during clicks, selection, and browsing steps performed by the user 'i' in the material 'j' correspondingly. $Max_{1\leq j\leq |M|}\left(LP_{ij}^c\right), Max_{1\leq j\leq |M|}\left(LP_{ij}^s\right), Max_{1\leq j\leq |M|}\left(LP_{ij}^l\right)$ describes the highest no. of clicks, selections, and browsings performed by the User 'i' in the material 'M'. $Min_{1\leq j\leq |M|}\left(LP_{ij}^c\right), Min_{1\leq j\leq |M|}(LP_{ij}^s)$, $Min_{1\leq j\leq |M|}\left(LP_{ij}^l\right)$ describes the lesser no. of clicks, selections, and browsing performed by the User 'i' in the material 'M'. The proposed personalized recommendation system is used to improve users' browsing knowledge.

4 Experimental Results

The experiment is deployed in Java with "Oracle Easy" as the Database. The performance chart of the proposed Enhanced DBSCAN and TSVM-based hybrid recommendation scheme is computed and distinguished from the existing state-of-the-art methods and from the previous methods. The present and proposed methods are distinguished with respect to query processing time, Mean absolute error (MAE) metric, and accuracy. The MOVIE LENS Data is used for an experimental purpose, which is open-sourced and available from the data world. The proposed method, DBSCN: TSVM, is then compared with other state-of-the-art methods, such as K-Means Neural Network (KNN), DBSCAN, Simple Support Vector Machine (SVM), Artificial Neural Network (ANN), Collaborative Filter Partial Timing (CFPT), and Extended Jacard Similarity (EJS).

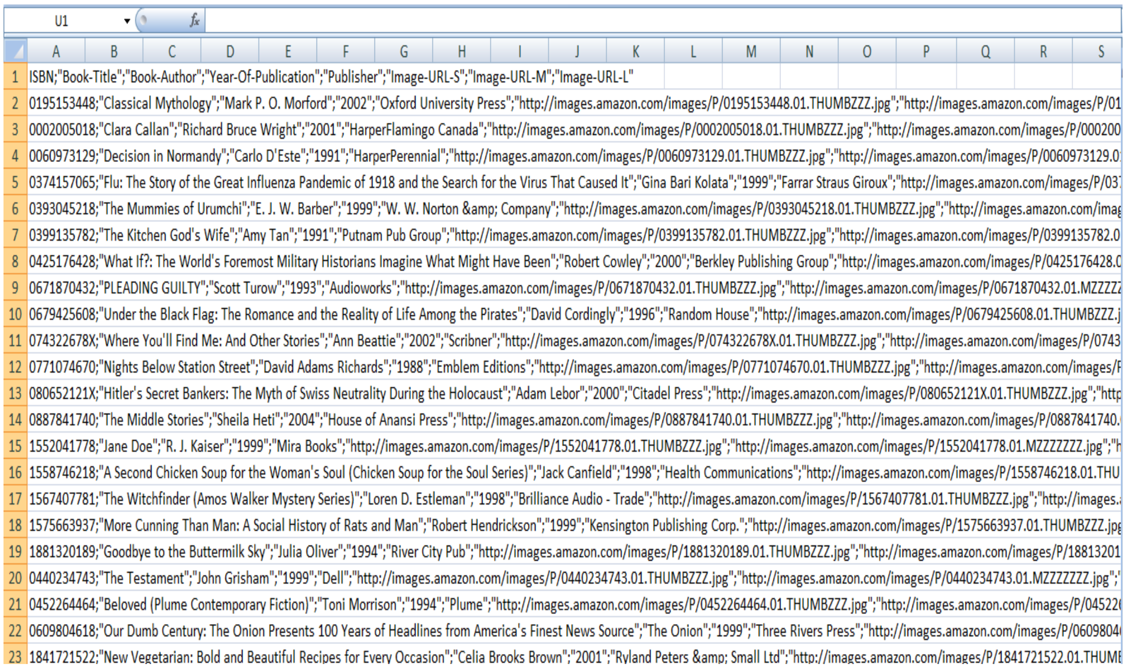

Figure 2: Dataset

The overall result analysis is shown in Table 4.1.

Table 1: Result analysis

Methods	Processing Time in Ms	Mean Absolute Error	Accuracy
KNN	0.92	0.96	0.69
DBSCAN	0.86	0.82	0.78
SVM	0.98	0.89	0.81
ANN	0.72	0.76	0.83
CFPT	0.68	0.62	0.79
EJS	0.86	0.73	0.80

DBSCAN: TSVM	0.41	0.25	0.96

4.1 Query Processing Time

The query processing time is defined as the time taken for the recommendation set to be generated for a particular click, selection, or search operation by the user. The experiment was conducted with 100 users as the initial set, which was slowly increased to 100000 users in a similar environment. The average time taken is calculated for each instance and is recorded. Figure 4.2 shows the comparative analysis of the query processing time of the proposed method with that of other state-of-the-art methods.

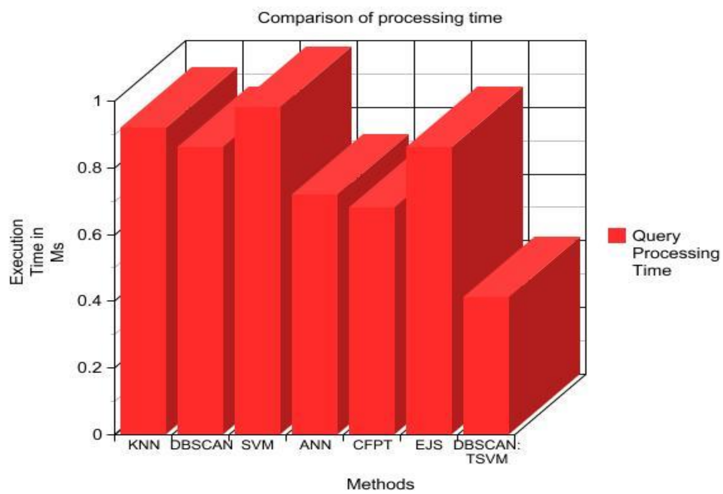

Figure 3: Comparison of Query Processing Time

Figure 3 represents the comparison chart of query processing time performance for the proposed Enhanced DBSCAN with TSVM-based hybrid recommendation and existing methods. The proposed system accomplished better output than the current system, which is proven in the experimental results.

4.2 Mean Absolute Error (MAE) metric

The MAE is figured by summing these total blunders of the N, which is proportional to ratings–forecast matches, and afterward calculating the normal esteem. For each combination, <pi, qi> of anticipated appraisals pi and the genuine evaluations qi, this metric yields the supreme blunder among them, i.e., | pi-qi |. To figure the measurable exactness, the MAE metric is used, which is a calculation of the suggestions' deviations from their actual user-indicated values. Formally,

$$\text{MAE}= \left(\sum_{i=1}^{N} |p_i - q_i|\right)/N \quad (7)$$

Figure 4 represents the comparison output of the proposed enhanced DBSCAN and TSVM-based hybrid recommendation, existing DBSCAN with SVM-based hybrid recommendation, and trust-based hybrid recommendation strategy with respect to MAE. The designed system accomplished better output than the current system, which is proven in the experimental results.

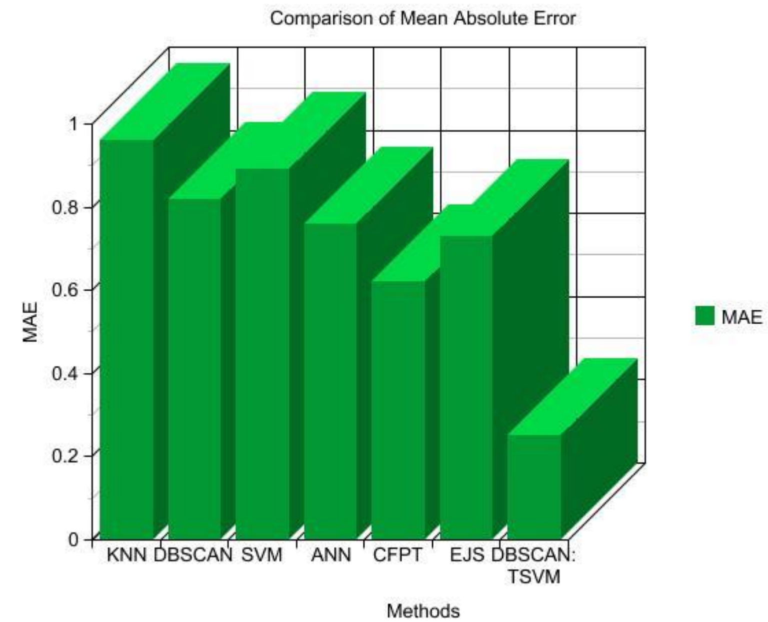

Figure 4: MAE comparison

4.3 Accuracy comparison

The accuracy is calculated by initially calculating the True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN), and through the calculation as shown below,

$$\text{Accuracy} = TP + TN / TP+TN+FP+FN \tag{8}$$

Figure 5 represents a comparison of the accuracy performance for the proposed Enhanced DBSCAN with TSVM-based hybrid recommendation and existing methods. It is found that the proposed method outperforms the existing methods with an accuracy of 96%.

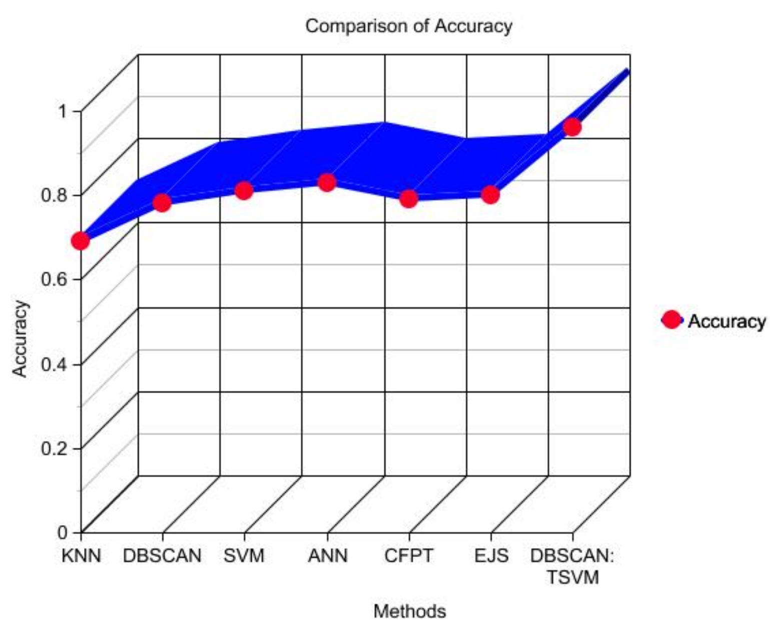

Figure 5: Accuracy comparison

5 Conclusion

The recommender system framework is provided for a personalized item recommendation, specifically giving the recommendations to help Users identify and choose the effective information. In our work, clustering is done by the Enhanced DBSCAN algorithm, which works according to the browsing style of the Users. Then the TSVM algorithm computes the Users' evaluation based on their habits and their interests. The student preferences and browsing behavior be able to be described by means of the subsequent three ways, like clicks, selections, and browsing. Finally, the experimental result shows that the designed system accomplishes good output when distinguished from the current system with respect to Processing time. MAE metric and accuracy.

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Conflicts of Interest: The authors declare no conflicts of interest to report regarding the present study.

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